

How Do We Build a Qubit?

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1 Introduction

A **qubit** (quantum bit) is the fundamental unit of quantum information. Unlike a classical bit which exists in state 0 or 1, a qubit exists in a **superposition** of both states.

Mathematically, a qubit is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

where:

- $\alpha, \beta \in \mathbb{C}$
- $|\alpha|^2 + |\beta|^2 = 1$

The challenge of building a qubit is to engineer a **physical quantum system** that behaves exactly like this two-level quantum system.

2 Mathematical Requirements of a Qubit

To physically realize a qubit, a system must:

1. Have exactly two distinguishable quantum states.
2. Allow coherent superposition.
3. Allow controlled unitary transformations.
4. Allow measurement in a chosen basis.
5. Maintain coherence long enough to compute.

Any physical system satisfying these properties can serve as a qubit.

3 General Physical Principle

A qubit is built from a **two-level quantum system**. Many physical systems naturally possess such two-level subspaces.

Common physical realizations include:

- Electron spin
- Photon polarization
- Superconducting circuits
- Trapped ions
- Quantum dots

We now explain how these are engineered.

4 Spin Qubits

4.1 Physical Basis

An electron has intrinsic angular momentum called **spin**.

$$|0\rangle = |\uparrow\rangle \quad (2)$$

$$|1\rangle = |\downarrow\rangle \quad (3)$$

The Hamiltonian in a magnetic field B is:

$$H = -\gamma \vec{B} \cdot \vec{S} \quad (4)$$

Energy splitting:

$$\Delta E = g\mu_B B \quad (5)$$

This creates a controllable two-level system.

4.2 Control

Using microwave pulses resonant at frequency:

$$\omega = \frac{\Delta E}{\hbar} \quad (6)$$

we induce Rabi oscillations:

$$P_{|1\rangle}(t) = \sin^2\left(\frac{\Omega t}{2}\right) \quad (7)$$

4.3 Measurement

Measured via spin-dependent tunneling or magnetic resonance detection.

5 Superconducting Qubits

This is currently the leading architecture in quantum processors.

5.1 Josephson Junction

A Josephson junction consists of two superconductors separated by a thin insulator.

The Josephson relations:

$$I = I_c \sin \phi \quad (8)$$

$$\frac{d\phi}{dt} = \frac{2eV}{\hbar} \quad (9)$$

where ϕ is the phase difference.

5.2 Nonlinear Oscillator

The Josephson junction creates a nonlinear inductance:

$$E_J(1 - \cos \phi) \tag{10}$$

The full Hamiltonian:

$$H = 4E_C n^2 - E_J \cos \phi \tag{11}$$

where:

- $E_C = \frac{e^2}{2C}$ (charging energy)
- E_J is Josephson energy
- n is Cooper pair number operator

Quantization yields discrete energy levels.

The lowest two levels are used as:

$$|0\rangle = \text{Ground state} \tag{12}$$

$$|1\rangle = \text{First excited state} \tag{13}$$

5.3 Example: Transmon Qubit

The transmon operates in regime:

$$\frac{E_J}{E_C} \gg 1 \tag{14}$$

This reduces charge noise sensitivity.

5.4 Control

Microwave pulses drive transitions between $|0\rangle$ and $|1\rangle$.

5.5 Measurement

Coupled to microwave resonator. State-dependent frequency shift enables dispersive readout.

6 Trapped Ion Qubits

6.1 Physical System

Single ion confined in electromagnetic trap.

Two internal atomic states serve as:

$$|0\rangle = |g\rangle \tag{15}$$

$$|1\rangle = |e\rangle \tag{16}$$

6.2 Hamiltonian

$$H = \frac{\hbar\omega_0}{2}\sigma_z \tag{17}$$

Laser pulses implement:

$$H_I = \hbar\Omega(\sigma^+e^{-i\omega t} + \sigma^-e^{i\omega t}) \tag{18}$$

6.3 Measurement

State-dependent fluorescence detection.

7 Photon Qubits

7.1 Polarization Encoding

$$|0\rangle = |H\rangle \tag{19}$$

$$|1\rangle = |V\rangle \tag{20}$$

General state:

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle \tag{21}$$

Controlled via waveplates and beam splitters.

8 Engineering Challenges

Building a qubit requires solving:

8.1 Decoherence

Environmental interaction causes:

$$\rho \rightarrow \mathcal{E}(\rho) \tag{22}$$

Characterized by:

- T_1 (relaxation time)
- T_2 (dephasing time)

8.2 Isolation vs Control Tradeoff

Strong coupling enables control but increases decoherence.

9 DiVincenzo Criteria

A physical implementation must satisfy:

1. Scalable qubits
2. Initialization capability
3. Long coherence times
4. Universal gate set
5. Qubit-specific measurement

10 Summary: What Does It Mean to Build a Qubit?

To build a qubit we:

1. Identify a controllable two-level quantum system.
2. Engineer energy splitting.
3. Enable coherent superposition.
4. Implement unitary control (quantum gates).
5. Implement reliable measurement.
6. Protect against decoherence.

Thus, a qubit is not merely a theoretical object, but a carefully engineered two-level quantum system that preserves coherence and allows precise quantum control.